① Seed selection: which $\mathbf{x}_i$ are seeds

SMOTE (2002)

$\mathbf{x}_i \sim \mathrm{Uniform}(S_{\min})$: every minority point, equal weight

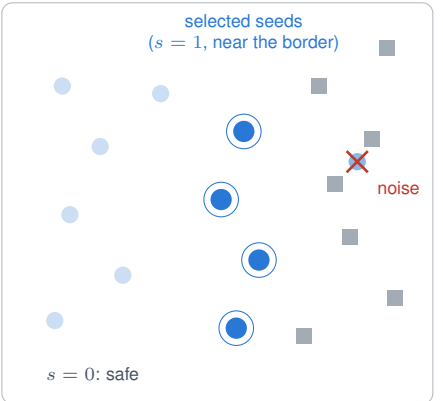
Generalised

$$\mathbf{x}_i \sim w(i) = \frac{s(\mathbf{x}_i)}{\sum_{k \in S_{\min}} s(\mathbf{x}_k)}, \quad s(\mathbf{x}_i) \geq 0$$

Example in the corpus

Borderline-SMOTE: $s(\mathbf{x}_i) = \mathbf{1}\left[\frac{k}{2} \leq m_i < k\right]$, m_i = majority points among the k nearest neighbours of $\mathbf{x}_i$

A score of zero removes a point as a seed. Border-focused scores favour points near the majority; safety scores (CISO, QC-SMOTE) favour dense, safe points.



② Sample allocation: how many points each seed gets

SMOTE (2002)

$g_i = N/100$ for every seed

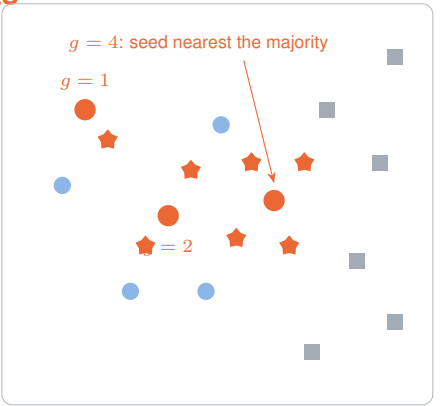
Generalised

$g_i = G \cdot \hat{r}_i, \quad \hat{r}_i = \dfrac{r_i}{\sum_k r_k}, \quad G = \sum_i g_i$ (total to generate)

Example in the corpus

ADASYN: $r_i = m_i/k$ (share of majority neighbours), $G = \big(|S_{\text{maj}}| - |S_{\text{min}}|\big)\,\beta$

Seeds in harder regions receive more points. Other methods set g_i per cluster (k-means SMOTE: by cluster sparsity) or per class (SCUT, GDHS: up to the mean class size).



③ Neighbour choice: which partner $\mathbf{x}_j$

SMOTE (2002)

$\mathbf{x}_j \sim \mathrm{Uniform}\big(N_5(\mathbf{x}_i)\big)$: 5 nearest minority neighbours, Euclidean

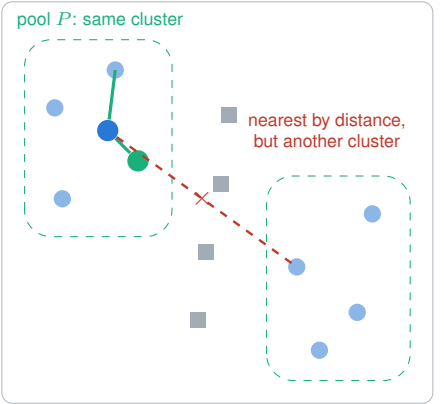
Generalised

$\mathbf{x}_j \sim q(j \mid i) \;\propto\; \omega_{ij}\,\mathbf{1}\big[\mathbf{x}_j \in P(\mathbf{x}_i)\big]$, neighbours under a metric d

Example in the corpus

Cluster-SMOTE: $P(\mathbf{x}_i)$ = neighbours in the same cluster as $\mathbf{x}_i$; Borderline-SMOTE2: P also holds the nearest majority point; FW-SMOTE: feature-weighted d

Three things can change: the candidate pool P, the weights ω_{ij} on the directions (SMOM, GDHS, CISO), and the distance d.



④ Placement rule: where $\mathbf{x}_{\text{new}}$ lands

SMOTE (2002)

$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i),\ \ \lambda \sim U(0,1)$: on the segment

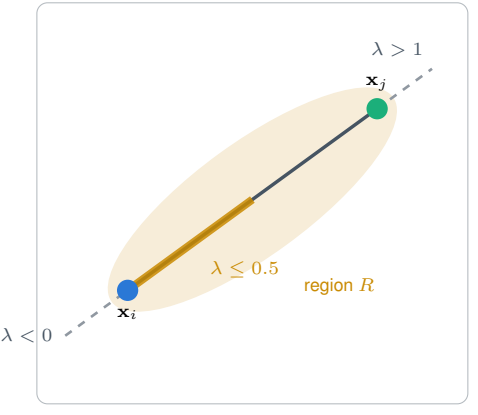
Generalised

$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i)$ with $\lambda \sim r(\lambda)$, or $\mathbf{x}_{\text{new}} \in R(\mathbf{x}_i, \mathbf{x}_j)$, or $\mathbf{x}_{\text{new}} = \sum_m \alpha_m \mathbf{x}_m$

Example in the corpus

Borderline-SMOTE2: $\lambda \sim U(0, 0.5)$; HDE: $\lambda \sim N(0,1)$; G-SMOTE: R = truncated, deformed hypersphere; GAMO: learned $\alpha_m \geq 0$, $\sum_m \alpha_m = 1$

A new law for λ keeps the line; a region R leaves it; λ outside $[0, 1]$ extrapolates (MKC-SMOTE, HDE); weights α_m combine more than two instances.



⑤ Post-filtering: what is kept after generation

SMOTE (2002)

$S' = S \cup S_{\text{syn}}$: every synthetic point is kept

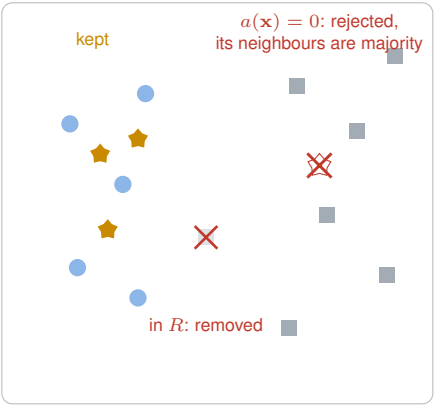
Generalised

$$S' = \Big(S \cup \{\mathbf{x} \in S_{\text{syn}} : a(\mathbf{x}) = 1\}\Big) \setminus R$$

Example in the corpus

KNNOR: $a(\mathbf{x}) = \mathbf{1}[\text{all } k \text{ nearest neighbours of } \mathbf{x} \text{ are minority}]$;
SMOTE-ENN: R = original or synthetic examples misclassified by their 3 nearest neighbours

a rejects or replaces candidates (KNNOR, SVDD-WSMOTE, QC-SMOTE); R cleans the enlarged set (SMOTE-Tomek, SMOTE-IPF) or removes majority points in the synthetic region (NROMM, GDHS).



⑥ Inside a pipeline: where generation runs

SMOTE (2002)

$h = \mathcal{A}\big(D \cup \mathrm{SMOTE}(D)\big)$: generate once, then train

Generalised

for $t = 1, \ldots, T$: $D_t = D \cup \mathrm{SMOTE}\big(D;\, \mathcal{D}_t\big)$, $h_t = \mathcal{A}(D_t)$;
$H(\mathbf{x}) = \arg\max_y \sum_t \alpha_t\, h_t(\mathbf{x}, y)$

Example in the corpus

SMOTEBoost: $\mathcal{D}_t$ = boosting weights of round t (AdaBoost.M2); the synthetic points are discarded after each round

Generation is repeated in every boosting round (SMOTEBoost, RAMOBoost, RBBoost), bagging member (HDE), wrapper step (LS) or training step (DSRBF, GAMO).

